Lee Jeong Hoon

M.S. Student in Computer Science

Programming Systems Lab, Hanyang University

Static Analysis · Code Intelligence · Robotics

Email: jeonghuni443@gmail.com

Mobile: +82 01-7552-7598

LinkedIn: https://www.linkedin.com/in/%EC%A0%95%ED%9B%88-%EC%9D%B4-7b7a793b8/

Hanyang University: https://www.hanyang.ac.kr/

Profile

I am an M.S. student in Computer Science at Hanyang University's Programming Systems Lab, researching Type-4 malicious code clone detection and static-analysis alarm ranking. I combine hands-on hardware and control experience from my B.S. in Robotics Engineering with contrastive representation learning to identify recurring vulnerabilities and semantically similar code without handcrafted features. My experience translating international robotics meetings and MOU discussions, along with teaching English, helps me communicate complex technical ideas clearly across teams.

Born: 2003.07.02

Location: Sangnok-gu, Ansan, Korea

Military: Exempt

Education

Master of Engineering | Hanyang University

Computer Science · Programming Systems Lab | 2025.03 - 2027.02

GPA: 4.36 / 4.5

Research in encoder-only LLM fine-tuning, contrastive learning, static analysis, code clone detection, and recurring vulnerability detection.

Bachelor of Engineering | Hanyang University ERICA

Robotics Engineering | 2022.03 - 2025.02

Project-focused study in mobile robotics, embedded systems, FPGA design, 3D CAD, and rapid prototyping.

Experience

Graduate Researcher

Programming Systems Lab, Hanyang University | 2025.03 - Present

Ansan, Korea

Researching Type-4 code clone detection and static-analysis alarm ranking with contrastively trained encoder-only LLMs. Work includes trace-text representation, sliding-window attention pooling, multi-loss aggregation, experimental design, and benchmark evaluation.

International Communications & Technical Translator

Auto Robotics Co., Ltd. | 2024.03 - 2025.03

Daegu · Hybrid

Interpreted video and in-person meetings and translated email communications for MOU discussions with international companies, bridging technical and business requirements in robotics contexts.

3D CAD Designer & English Meeting Support

Siliconus Co., Ltd. | 2023.03 - 2023.09

Bucheon, Korea

Designed prototypes in SolidWorks and validated manufacturability through 3D printing. Supported technical communication in face-to-face meetings with international stakeholders.

English Translator & Interpreter

Auto Robotics Co., Ltd. | 2021.12 - 2023.01

Daegu, Korea

Provided Korean-English interpretation and translation for online and in-person MOU meetings and business email correspondence.

Projects

Trace-Based Contrastive Learning for Static Analysis Alarm Ranking Without Feature Engineering

Graduate Research | 2025.08 - 2026.06

Programming Systems Lab

Role: Trace Representation · Model Fine-tuning · Evaluation

Transforms source-to-sink traces into text and contrastively trains an encoder-only LLM to produce embeddings. New alarms are ranked by cosine similarity to confirmed vulnerability traces stored in SignatureDB.

- Evaluated ranking performance with a Juliet-derived SignatureDB and Debian alarms.

- Achieved 0.691 AUC after trace-text contrastive training, outperforming the prior approach.

Technologies: Static Analysis, Contrastive Learning, Encoder-only LLM, Python

Cross-language Type-4 Clone Detection with Sliding Window and Attention-enhanced Contrastive Learning

Graduate Research | 2025.03 - 2025.12

Programming Systems Lab

Role: Contrastive Pipeline · Long-code Representation · Experiments

Uses LLM-generated code descriptions as semantic guidance and combines sliding windows with attention pooling to overcome encoder length limits for cross-language Type-4 clone detection.

- Fused code-to-code, code-to-description, and cross code-to-description contrastive losses.

- Adaptively learned loss weights with uncertainty-weighted aggregation.

Technologies: Code Clone Detection, Attention Pooling, Multi-loss Learning, LLM

D-STAR: Domain-Specific Teaching Assistant Regex Subset Synthesizer

Software Research Project | 2026.03 - 2026.06

Hanyang University

Role: Synthesis Algorithm · DSL Search · Explainable Demo

A teaching-assistant-oriented synthesizer that learns validation regexes from positive and negative examples such as student IDs, names, emails, and assignment filenames.

- Pruned impossible candidates early to reduce the search space.

- Identified offending negative fragments and resynthesized with partial constraints.

Technologies: Program Synthesis, Regex, DSL, Python

Thrust-assisted Mobile Robot for Free Movement on Vertical Surfaces

Capstone Design | 2024.03 - 2024.12

Hanyang University ERICA

Role: Mechanical Design · Motion Control · Prototyping

Designed a modular elevated-work mobile robot that offsets gravity with duct-fan thrust and moves omnidirectionally across vertical surfaces using omni wheels.

- Derived adhesion and motion conditions from thrust, friction, and center-of-mass analysis.

- Completed 3D CAD, fabrication, hardware integration, and mobility testing.

- Capstone design output: Adhesion and Omnidirectional Locomotion System for a Thrust-assisted Wall-climbing Mobile Robot

Technologies: Robotics, Raspberry Pi, 3D CAD, Control

Kevlar Tendon-driven Five-finger Robotic Gripper

Robotics Project | 2023.09 - 2023.12

Hanyang University ERICA

Role: Mechanism Design · Fabrication · Grasp Testing

Built a five-finger gripper actuated by Kevlar tendons and small pulleys. Silicone tips and webbed mesh distribute contact force for fragile or deformable objects.

- Coupled multiple fingers through a compact tendon path without complex actuators.

- Improved grasp stability with silicone contact surfaces and flexible mesh.

Technologies: Mechanical Design, 3D Printing, SolidWorks, Prototyping

Dual Cooperative Elevator Simulator on Altera DE2-115

Digital Systems | 2022.09 - 2022.12

Hanyang University ERICA

Role: State-machine Design · FPGA Implementation · Verification

An FPGA simulator coordinating two elevators so the closer car answers each floor call, with LEDs representing motion and door state.

- Modeled call assignment, direction changes, and door operation as finite-state machines.

- Implemented an overload event and switch-based recovery scenario.

Technologies: FPGA, Verilog, FSM, Altera DE2-115

Publications

Trace-Based Contrastive Learning for Static Analysis Alarm Ranking Without Feature Engineering

2026 | Conference Paper

Korea Computer Congress 2026 (KCC 2026)

Lee Jeong Hoon, Wooseok Lee

Outstanding Presentation Paper Award

Cross-language Type-4 Clone Detection with Sliding Window and Attention-enhanced Contrastive Learning

2025 | Research Poster

SIGPL Poster Session

Hao Wang, Lee Jeong Hoon, Wooseok Lee

Awards

Outstanding Presentation Paper Award

July 31, 2026 | Korean Institute of Information Scientists and Engineers (KIISE)

Korea Computer Congress 2026 (KCC 2026)

Track: Programming Languages | Certificate: No. 26-791

Trace-Based Contrastive Learning for Static Analysis Alarm Ranking Without Feature Engineering

Awarded by the president of KIISE after the paper was selected as an outstanding presentation paper in the Programming Languages track of Korea Computer Congress 2026.

Skills & Languages

Research & AI

Static Analysis · Contrastive Learning · Encoder-only LLM Fine-tuning · Code Clone Detection · Representation Learning · Program Synthesis

Programming

Python · C / C++ · Java · JavaScript · Verilog · Git · Linux

Robotics & Fabrication

Raspberry Pi · SolidWorks · 3D CAD · 3D Printing · Embedded Systems · FPGA · Rapid Prototyping

Language & Communication

Korean · Native · English · Professional / Technical Translation · G-TELP Level 2 · Mastery · 98% (2026-09-08) — Listening 96%, Reading & Vocabulary 100%, Grammar 96% · TESL-based English Teaching · International Meeting Facilitation